

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12414497

Kind Code

B2

Date of Patent

September 16, 2025

Inventor(s)

Kinney; Colter W. et al.

Work machines incorporating fertilizer opener assemblies, fertilizer opener assemblies therefor, and methods of operating work machines

Abstract

Work machines, fertilizer opener assemblies for work machines, and methods of operating work machines are disclosed. A work machine includes a chassis, a frame, and a fertilizer opener assembly. The frame is coupled to the chassis for movement relative to the chassis. The fertilizer opener assembly is coupled to the frame to selectively form trenches in the soil for the deposition of fertilizer. The fertilizer opener assembly includes an actuator that is extendable and retractable to vary a length of the actuator and a cutting implement coupled to the actuator. Operation of the actuator drives movement of the cutting implement between a cutting position, in which the cutting implement makes contact with the soil to form trenches in the soil, and a transport position, in which the cutting implement is raised above the soil.

Inventors: Kinney; Colter W. (Lohrville, IA), Soto; Carlos Alberto (Gudalupe, MX), Wackerle; Bradley A. (Delmar, IA), Maro; Randall A. (Davenport, IA)

Applicant: Deere & Company (Moline, IL)

Family ID: 1000008820490

Assignee: Deere & Company (Moline, IL)

Appl. No.: 17/388194

Filed: July 29, 2021

Prior Publication Data

Document Identifier

US 20230036576 A1

Publication Date

Feb. 02, 2023

Publication Classification

Int. Cl.: **A01C7/20** (20060101); **A01B63/111** (20060101); **A01B69/00** (20060101); **A01C5/06** (20060101); **A01C15/00** (20060101)

U.S. Cl.:

CPC **A01C7/203** (20130101); **A01B63/111** (20130101); **A01B69/00** (20130101); **A01C5/064** (20130101); **A01C15/001** (20130101); A01C15/005 (20130101)

Field of Classification Search

CPC: A01C (7/203); A01C (5/064); A01C (15/001); A01C (15/005); A01C (7/06); A01C (7/201); A01C (7/205); A01B (63/111); A01B (69/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4796550	12/1988	Van Natta et al.	N/A	N/A
7261048	12/2006	Hantke	N/A	N/A
7581503	12/2008	Martin et al.	N/A	N/A
2010/0230124	12/2009	Martin	N/A	N/A
2015/0264857	12/2014	Achen	172/260.5	A01B 63/32
2017/0215334	12/2016	Dienst et al.	N/A	N/A
2021/0059099	12/2020	Menezes	N/A	A01B 73/00

OTHER PUBLICATIONS

Extended European Search Report and Written Opinion issued in European Patent Application No. 22182122.6, dated Dec. 12, 2022, in 09 pages. cited by applicant

U.S. Appl. No. 16/402,803. cited by applicant

Primary Examiner: Zarroli; Michael C

Attorney, Agent or Firm: Taft Stettinius & Hollister LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates, generally, to work machines such as agricultural planters, for example, and, more specifically, to fertilizer opener assemblies for work machines.

BACKGROUND

(2) Some work machines, such as agricultural planters, seeding machines, and/or grain drills, for example, form trenches in soil for the deposition of fertilizer and/or seed. Those work machines may include a cutting implement that contacts the soil to form the trenches. It may be desirable to maintain the cutting implement in contact with the soil at a particular depth. Additionally, it may be desirable to move the cutting implement independently of a toolbar or frame to which the cutting implement is attached. Furthermore, it may be desirable to maintain an orientation of the cutting implement relative to the soil. Accordingly, devices, systems, and/or methods that address the aforementioned concerns, among other things, remain an area of interest.

SUMMARY

(3) The present disclosure may comprise one or more of the following features and combinations thereof.

(4) According to one aspect of the present disclosure, a work machine may include a chassis, a frame coupled to the chassis for movement relative thereto, and a fertilizer opener assembly coupled to the frame to selectively form trenches in the soil for the deposition of fertilizer. The fertilizer opener assembly may include an actuator that is extendable and retractable to vary a length of the actuator and a cutting implement coupled to the actuator. Operation of the actuator may drive movement of the cutting implement between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil. Operation of the actuator may drive movement of the cutting implement between the cutting position and the transport position independently of movement of the frame relative to the chassis.

(5) In some embodiments, extension of the actuator may drive movement of the cutting implement toward the transport position, and retraction of the actuator may drive movement of the cutting implement toward the cutting position. Additionally, in some embodiments, the work machine may include a control system coupled to the chassis that includes at least one sensor to provide sensor data indicative of a force applied by the actuator to the cutting implement and a controller communicatively coupled to the at least one sensor, the controller may include memory having instructions stored therein and a processor coupled to the memory, and the instructions may be executable by the processor to cause the processor to receive the sensor data from the at least one sensor and to control operation of the actuator based on the sensor data such that a relatively constant force is applied by the actuator to the cutting implement when the cutting implement is in the cutting position. The instructions may be executable by the processor to cause the processor to selectively retract the actuator to move the cutting implement into the soil in a first direction when the cutting implement is in the cutting position, and the instructions may be executable by the processor to cause the processor to selectively extend the actuator to move the cutting implement in a second direction opposite the first direction when the cutting implement is in the cutting position.

(6) In some embodiments, the fertilizer opener assembly may include a positional adjustment linkage coupled to the actuator and the cutting implement such that operation of the actuator drives movement of the cutting implement through the positional adjustment linkage, and the positional adjustment linkage may include a first positional adjustment arm that is directly connected to the actuator and coupled to the cutting implement and a second positional adjustment arm spaced from the first positional arm that is coupled to the actuator and to the cutting implement. The positional adjustment linkage may include a cutting implement guide link coupled to the cutting implement to at least partially define an angular orientation of the cutting implement relative to the soil, the cutting implement guide link may include a first face and a second face arranged opposite the first face, and the first positional adjustment arm may contact each of the first and second faces of the cutting implement guide link and the second positional adjustment arm may contact the second face of the cutting implement guide link. Additionally, in some embodiments, the fertilizer opener assembly may include a mount kit to attach the fertilizer opener assembly to the frame, the mount kit may include a mount plate to contact the frame, a mount bracket extending outwardly from the mount plate that is affixed to the mount plate and to the actuator, a first mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm and the second positional adjustment arm at another end thereof opposite the one end, and a second mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm, the second positional adjustment arm, and the first mount link at another end thereof opposite the one end, and the first and second positional adjustment arms may be pivotally coupled to the first and second mount links to permit movement of the cutting implement between the cutting position and the transport position.

(7) In some embodiments, during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link may at least partially cooperate to maintain a relatively constant angular orientation of the cutting implement relative to the soil. Additionally, in some embodiments, during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link may at least partially cooperate to maintain a relatively constant upright orientation of the cutting implement guide link in which the cutting implement guide link extends away from the soil in a vertical direction. Further, in some embodiments still, during movement of the cutting implement between the cutting position and the transport position, the first and second positional adjustment arms may be arranged to extend parallel to one another.

(8) According to another aspect of the present disclosure, a fertilizer opener assembly for a work machine including a chassis and a frame coupled to the chassis for movement relative thereto may include an actuator, a cutting implement to selectively form trenches in the soil for the deposition of fertilizer, and a positional adjustment linkage. The actuator may be extendable and retractable to vary a length of the actuator. The cutting implement may be movable relative to the soil between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil. The positional adjustment linkage may be coupled to the actuator and the cutting implement such that operation of the actuator drives movement of the cutting implement between the cutting position and the transport position through the positional adjustment linkage. Extension of the actuator may drive movement of the cutting implement through the positional adjustment linkage toward the transport position, and retraction of the actuator may drive movement of the cutting implement through the positional adjustment linkage toward the cutting position.

(9) In some embodiments, the positional adjustment linkage may include a first positional adjustment arm that is directly connected to the actuator and coupled to the cutting implement and a second positional adjustment arm spaced from the first positional arm that is coupled to the actuator and to the cutting implement, and during movement of the cutting implement between the cutting position and the transport position, the first and second positional adjustment arms may be arranged to extend parallel to one another. Additionally, in some embodiments, a control system may include at least one sensor to provide sensor data indicative of a force applied to the cutting implement by the actuator and a controller communicatively coupled to the at least one sensor, the controller may include memory having instructions stored therein and a processor coupled to the memory, and the instructions may be executable by the processor to cause the processor to receive the sensor data from the at least one sensor and to control operation of the actuator based on the sensor data such that a relatively constant force is applied to the cutting implement by the actuator when the cutting implement is in the cutting position.

(10) In some embodiments, the positional adjustment linkage may include a cutting implement guide link coupled to the cutting implement to at least partially define an angular orientation of the cutting implement relative to the soil, the cutting implement guide link may include a first face and a second face arranged opposite the first face, and the first positional adjustment arm may contact each of the first and second faces of the cutting implement guide link and the second positional adjustment arm may contact the second face of the cutting implement guide link. During movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link may at least partially cooperate to maintain a relatively constant angular orientation of the cutting implement relative to the soil. During movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link may at least partially cooperate to maintain a relatively constant upright orientation of the cutting implement guide link in which the cutting

implement guide link extends away from the soil in a vertical direction. Additionally, in some embodiments, the fertilizer opener assembly may include a mount kit to attach the fertilizer opener assembly to the frame, the mount kit may include a mount plate to contact the frame, a mount bracket extending outwardly from the mount plate that is affixed to the mount plate and to the actuator, a first mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm and the second positional adjustment arm at another end thereof opposite the one end, and a second mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm, the second positional adjustment arm, and the first mount link at another end thereof opposite the one end, and the first and second positional adjustment arms may be pivotally coupled to the first and second mount links to permit movement of the cutting implement between the cutting position and the transport position.

(11) According to yet another aspect of the present disclosure, a method of operating a work machine including a chassis, a frame coupled to the chassis for movement relative thereto, and a fertilizer opener assembly coupled to the frame to selectively form trenches in the soil for the deposition of fertilizer may include directing, by a controller, movement of the frame relative to the chassis, and directing, by the controller, movement of a cutting implement of the fertilizer opener assembly independently of movement of the frame relative to the chassis. Directing movement of the cutting implement independently of movement of the frame relative to the chassis may include providing, by the controller, control signals to an actuator of the fertilizer opener assembly to cause extension or retraction of the actuator such that the cutting implement moves between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil.

(12) In some embodiments, the method may include directing, by the controller, movement of the cutting implement to the cutting position independently of movement of the frame relative to the chassis, determining, by the controller, a depth of the cutting implement in the soil when the cutting implement is in the cutting position, comparing, by the controller, the determined depth to a reference depth, and adjusting, by the controller, a position of the cutting implement based on the comparison. The method may include determining, by the controller, a force applied to the cutting implement by the actuator when the cutting implement is in the cutting position, comparing, by the controller, the determined force to a reference force, and adjusting, by the controller, movement of the actuator based on the comparison such that a relatively constant force is applied by the actuator to the cutting implement when the cutting implement is in the cutting position.

(13) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention described herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

(2) FIG. 1 is a side elevation view of a work machine including a fertilizer opener assembly that is depicted diagrammatically for the sake of simplicity;

(3) FIG. 2 is a perspective view of one embodiment of the fertilizer opener assembly depicted in FIG. 1 with a cutting implement of the fertilizer opener system in a lowered cutting position;

(4) FIG. 3 is a perspective view of the fertilizer opener assembly shown in FIG. 2 with the cutting implement in a raised transport position;

- (5) FIG. 4 is a front view of the fertilizer opener assembly shown in FIG. 2 with the cutting implement in the lowered cutting position;
- (6) FIG. 5 is a rear view of the fertilizer opener assembly shown in FIG. 2 with the cutting implement in the lowered cutting position;
- (7) FIG. 6 is a diagrammatic view of a control system that may be included in the work machine of FIG. 1;
- (8) FIG. 7 is a diagrammatic view of a number of modules that may be included in a controller of the control system shown in FIG. 6;
- (9) FIG. 8 is a simplified flowchart of one portion of a method of operating the work machine that may be performed by the controller of the control system shown in FIG. 6; and
- (10) FIG. 9 is a simplified flowchart of another portion of the method of FIG. 8.

DETAILED DESCRIPTION

(11) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

(12) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).

(13) In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

(14) A number of features described below may be illustrated in the drawings in phantom. Depiction of certain features in phantom is intended to convey that those features may be hidden or present in one or more embodiments, while not necessarily present in other embodiments. Additionally, in the one or more embodiments in which those features may be present, illustration of the features in phantom is intended to convey that the features may have location(s) and/or position(s) different from the locations(s) and/or position(s) shown.

(15) Referring now to FIG. 1, a work machine **100** is shown. The work machine **100** may include a main frame or chassis **104** to which a tow bar **102** is attached to allow the work machine **100** to be coupled to an agricultural machine, such as a tractor (not shown), for example. The work machine **100** may include one or more wheels **106** for traversing along the soil or ground, as the case may be.

(16) In the illustrative embodiment, the work machine **100** is embodied as, or otherwise includes, a planter or planting implement. However, in other embodiments, it should be appreciated that the work machine **100** may be embodied as, or otherwise include, other equipment adapted for use in a

variety of applications. For example, in some embodiments, the work machine **100** of the present disclosure may be embodied as, included in, or otherwise adapted for use with, front end loaders, scraper systems, cutters and shredders, hay and forage equipment, seeding equipment, sprayers and applicators, tillage equipment, utility vehicles, mowers, dump trucks, backhoes, track loaders, crawler loaders, dozers, excavators, motor graders, skid steers, tractor loaders, wheel loaders, rakes, aerators, skidders, bunchers, forwarders, harvesters, swing machines, knuckleboom loaders, diesel engines, axles, planetary gear drives, pump drives, transmissions, generators, or marine engines, among other suitable equipment.

(17) The planter **100** may include a plurality of planting units **118** attached to the main frame **104**, at least in some embodiments. As shown in FIG. 1, a planting unit **118** is attached to the main frame **104** via a parallel linkage **116**. The parallel linkage **116** may allow for independent vertical movement of the planting unit **118** relative to the chassis **104** as it traverses along uneven ground in use of the planter **100**, among other things.

(18) Each planting unit **118** may include its own support structure **128** to which a seed hopper **120** is coupled, at least in some embodiments. The seed hopper **120** may store seed to be planted by the planting unit **118** during a planting operation. Seed may be deposited within a trench or furrow formed by a trench-forming disk **122**. The depth at which the trench-forming disk **122** is positioned relative to the soil may be set by a gauge wheel **124** and a depth-setting mechanism (not shown). A pair of closing wheels **126** may be coupled to the support structure **128** of the planting unit **118** in order to close or cover the trench with soil.

(19) In the illustrative embodiment, a frame **110** is coupled to, and capable of movement relative to, the chassis **104**. In some embodiments, the frame **110** may be embodied as, or otherwise include, a toolbar or drawbar. In any case, at least one fertilizer opener assembly **210** (shown in detail in FIG. 2) is coupled to the frame **110** to selectively form trenches in the soil for the deposition of fertilizer. A hopper or tank **108** may be used to store fertilizer (e.g., dry granular fertilizer or gaseous or liquid fertilizer) for deposition in the trenches formed by the fertilizer opener assembly **210**. As discussed below, each fertilizer opener assembly **210** includes a cutting disk **212** and a gauge wheel **214**, among other things. The cutting disk **212** may be set and/or positioned at a defined depth into the soil to form a furrow or trench therein. The gauge wheel **214** may be used to set the depth of the furrow or trench to be formed by the cutting disk **212**. For the purposes of the present disclosure, the cutting disk **212** and the gauge wheel **214** may be collectively referred to as, or otherwise included as components of, a cutting implement.

(20) Referring now to FIG. 2, the illustrative fertilizer opener assembly **210** includes an actuator **220** (e.g., a double-acting hydraulic cylinder, an electro-hydraulic actuator, a linear actuator, a pneumatic actuator, or the like) that is extendable and retractable to vary a length *L* of the actuator **220**. The cutting disk **212** is illustratively coupled to the actuator **220** through a positional adjustment linkage **230**, as described in greater detail below. In the illustrative embodiment, operation of the actuator **220** drives movement of the cutting disk **212** between a cutting position **200** and transport position **300** (see FIG. 3). In the cutting position **200**, the cutting disk **212** makes contact with the soil to form trenches therein, whereas in the transport position **300**, the cutting disk **212** is raised above the soil and does not contact the soil. In the illustrative embodiment, operation of the actuator **220** drives movement of the cutting disk **212** between the cutting position **200** and the transport position **300** independently of movement of the frame **110** relative to the chassis **104**. Accordingly, positional adjustment of the fertilizer opener assembly **210** may be performed independently of movement of the frame **110** relative to the chassis **104**.

(21) As best seen in FIG. 2, retraction of the actuator **220** (or movement of the actuator **220** to a retracted state) drives movement of the cutting disk **212** toward the cutting position **200**. As best seen in FIG. 3, extension of the actuator **220** (or movement of the actuator **220** to an extended state) drives movement of the cutting disk **212** toward the transport position **300**. In the illustrative embodiment, the positional adjustment linkage **230** is coupled to the actuator **220** and the cutting

disk **212** such that operation of the actuator **220** drives movement of the cutting disk **212** between the cutting position **200** and the transport position **300** through the positional adjustment linkage **230**. Therefore, extension of the actuator **220** drives movement of the cutting disk **212** through the positional adjustment linkage **230** toward the transport position **300**, whereas retraction of the actuator **220** drives movement of the cutting disk **212** through the positional adjustment linkage **230** toward the cutting position **200**.

(22) Referring now to FIGS. 2 and 3, the positional adjustment linkage **230** includes positional adjustment arms **240**, **250**. The positional adjustment arm **240** includes a mount flange **242** that defines an end **244** of the arm **240** arranged opposite an end **246** of the arm **240**. The mount flange **242** is directly connected to the actuator **220** and the end **246** of the arm **240** is directly connected to a cutting implement guide link **260** included in the positional adjustment linkage **230** that is coupled to the cutting disk **212**. The positional adjustment arm **250** is spaced from the positional adjustment arm **240** and extends from an end **252** to an end **254** arranged opposite the end **254**. The end **252** of the arm **250** is coupled to the actuator **220** and the end **254** of the arm **250** is coupled to the cutting implement guide link **260**.

(23) In the illustrative embodiment, the cutting implement guide link **260** is coupled to the cutting disk **212** such that the cutting implement guide link **260** at least partially defines an angular orientation α of the cutting disk **212** relative to the soil. The illustrative cutting implement guide link **260** is integrally formed with, includes, or is otherwise coupled to, a housing **262** affixed to a face **213** of the cutting disk **212**. The housing **262** supports an axle **264** for rotation about an axis A such that the cutting implement guide link **260** and the cutting disk **212** are at least partially configured for rotation about the axis A to adjust the angular orientation α of the cutting disk **212**, at least in some embodiments.

(24) As best seen in FIGS. 4 and 5, the illustrative cutting implement guide link **260** includes faces **462**, **464** that are arranged opposite one another. The end **246** of the positional adjustment arm **240** is defined by, otherwise includes, extensions **446**, **546** that are spaced apart from one another to define a notch **558** therebetween. The cutting implement guide link **260** is received in the notch **558** such the extensions **446**, **546** contact respective faces **462**, **464** of the guide link **260**. Additionally, when the cutting implement guide link **260** is received in the notch **558** as shown in FIGS. 4 and 5, the end **254** of the positional adjustment arm **250** contacts the face **464** of the guide link **260**. In the illustrative mounting arrangement, the positional adjustment arms **240**, **250** are pivotally coupled to the cutting implement guide link **260** adjacent their respective ends **246**, **254**.

(25) Referring now to FIGS. 2-5, in the illustrative embodiment, the fertilizer opener assembly **210** includes a mount kit **270** to attach the assembly **210** to the frame **110**. The illustrative mount kit **270** includes a mount plate **272** configured to contact the frame **110**, a mount bracket **274** extending outwardly from the mount plate **272** that is affixed to the mount plate **272** and the actuator **220**, a mount link **280**, and a mount link **286**. The mount link **280** is affixed to the mount bracket **274** at an end **282** thereof and coupled to the positional adjustment arms **240**, **250** at an end **284** thereof opposite the end **282**. The mount link **286** is affixed to the mount bracket **274** at an end **288** thereof and coupled to the positional adjustment arms **240**, **250** and the mount link **280** at end **290** thereof opposite the end **288**.

(26) As best seen in FIG. 4, the end **282** of the mount link **280** is affixed to a face **474** of the mount bracket **274** and the end **288** of the mount link **286** is integrally formed with, or otherwise coupled to, a face **476** of the mount bracket **274** arranged opposite the face **474**. The actuator **220** is pivotally coupled to a portion **478** of the mount bracket **274** and to the mount flange **242** of the arm **240**. A first fastener **490** extends through the mount link **280** (i.e., at the end **284**), the mount flange **242** of the positional adjustment arm **240**, and the mount link **286** (i.e., at the end **290**). A second fastener **492** spaced from the first fastener **490** extends through the mount link **280** (i.e., at the end **284**), the positional adjustment arm **250** (i.e., at the end **252**), and the mount link **286** (i.e., at the end **290**). As such, the positional adjustment arms **240**, **250** are pivotally coupled to the mount links

280, 286 by the fasteners **490, 492** to permit movement of the cutting disk **212** between the cutting and transport positions **200, 300**.

(27) The illustrative fertilizer opener assembly **210** includes a boot or scraper **216** arranged in confronting relation to the face **213** of the cutting disk **212**. In some embodiments, the boot or scraper **216** may be coupled to, and supported by, a portion of the cutting implement guide link **260**, such as the housing **262**, for example. In other embodiments, however, the boot or scraper **216** may be supported by a structure (not shown) mounted to the frame **110** and/or the chassis **104**. In any case, at least in some embodiments, the boot or scraper **216** may be positioned behind at least a portion of the cutting disk **212** (e.g., a forward edge or side thereof) relative to a forward or working direction. When the cutting disk **212** forms a trench in the soil as the fertilizer opener assembly **210** moves in the forward or working direction, the boot or scraper **216** may lie within the trench profile so as not to widen the trench formed in the soil by the disk **212**. In some embodiments, depending on the application, the boot or scraper **216** may be slightly offset from the disk **212** to widen the trench (e.g., by about ¼ inch) formed by the cutting disk **212**. It should be appreciated that the boot or scraper **216** has a contour closely matched with the trench (e.g., forward) side of the cutting disk **212** to effectively scrape mud, soil, and other debris from the trench side of the disk **212** in use thereof. Furthermore, it should be appreciated that the bottom portion of the boot or scraper **216** may be displaced into the soil to a desired depth at which fertilizer may be applied.

(28) Returning to FIGS. 2 and 3, to deploy the cutting disk **212** in the cutting position **200** in use of the assembly **210**, the actuator **220** is retracted as indicated above. In some embodiments, prior to retraction of the actuator **220**, the gauge wheel **214** may be used to set a desired penetration depth of the cutting disk **212** into the soil, at least in some embodiments. However, in other embodiments, the gauge wheel **214** may be used to set a desired penetration depth of the cutting disk **212** into the soil concurrently with, or following, retraction of the actuator **220**. In any case, retraction of the actuator **220** corresponds to, or is otherwise associated with, movement of the cutting disk **212** away from the transport position **300** toward the cutting position **200**. Conversely, extension of the actuator **220** to stow the cutting disk **212** in use of the assembly **210** corresponds to, or is otherwise associated with, movement of the cutting disk **212** away from the cutting position **200** toward the transport position **300**.

(29) In the illustrative embodiment, during movement of the cutting disk **212** between the cutting position **200** and the transport position **300**, the positional adjustment arms **240, 250** and the cutting implement guide link **260** of the positional adjustment linkage **230** at least partially cooperate to maintain a relatively constant angular orientation α of the cutting disk **212** relative to the soil. Additionally, during movement of the cutting disk **212** between the positions **200, 300**, the arms **240, 250** and the guide link **260** at least partially cooperate to maintain a relatively constant upright orientation β of the guide link **260** relative to the soil. In the relatively constant upright orientation β of the cutting implement guide link **260**, the guide link **260** extends away from the soil in a vertical direction indicated by arrow V.

(30) In the illustrative embodiment, during movement of the cutting disk **212** between the cutting position **200** and the transport position **300**, the positional adjustment arms **240, 250** are arranged to extend parallel to one another. The arrangement of the arms **240, 250** parallel to one another is maintained due at least in part to the connection of the arms **240, 250** to opposite faces **462, 464** of the cutting implement guide link **260**. Accordingly, the arms **240, 250** extend parallel to one another when the cutting disk **212** is raised or lowered relative to the soil.

(31) Referring now to FIG. 6, the illustrative planter **100** includes a control system **600** configured to control operation of the planter **100**. In some embodiments, the control system **600** may be incorporated into, dedicated to, and or otherwise adapted for use with, the fertilizer opener assembly **210**. In any case, the illustrative control system includes a controller **602**, one or more frame actuator(s) **610**, one or more frame sensor(s) **612**, one or more user input device(s) **614**, one

or more cutting implement position sensor(s) **616**, one or more cutting implement force measurement sensor(s) **618**, one or more cutting implement actuator valve(s) **620**, one or more cutting implement actuator sensor(s) **622**, a dashboard **624**, and one or more gauge wheel sensor(s) **630**. Each of the devices **610**, **612**, **614**, **616**, **618**, **620**, **622**, **624**, **630** is communicatively coupled to the controller **602**. The controller **602** includes a memory device **604** (or multiple memory devices) and a processor **606** communicatively coupled to the memory device **604**.

(32) The memory device **604** of the illustrative controller **602** may be embodied as any type of volatile (e.g., dynamic random access memory (DRAM), etc.) or non-volatile memory capable of storing data therein. Volatile memory may be embodied as a storage medium that requires power to maintain the state of data stored by the medium. Non-limiting examples of volatile memory may include various types of random access memory (RAM), such as dynamic random access memory (DRAM) or static random access memory (SRAM). One particular type of DRAM that may be used in a memory module is synchronous dynamic random access memory (SDRAM). In particular embodiments, DRAM of a memory component may comply with a standard promulgated by JEDEC, such as JESD79F for DDR SDRAM, JESD79-2F for DDR2 SDRAM, JESD79-3F for DDR3 SDRAM, JESD79-4A for DDR4 SDRAM, JESD209 for Low Power DDR (LPDDR), JESD209-2 for LPDDR2, JESD209-3 for LPDDR3, and JESD209-4 for LPDDR4 (these standards are available at www.jedec.org). Such standards (and similar standards) may be referred to as DDR-based standards and communication interfaces of the storage devices that implement such standards may be referred to as DDR-based interfaces.

(33) In some embodiments, the memory device **604** may be embodied as a block addressable memory, such as those based on NAND or NOR technologies. The memory device **604** may also include future generation nonvolatile devices, such as a three dimensional crosspoint memory device (e.g., Intel 3D XPoint™ memory), or other byte addressable write-in-place nonvolatile memory devices. In some embodiments, the memory device **604** may be embodied as, or may otherwise include, chalcogenide glass, multi-threshold level NAND flash memory, NOR flash memory, single or multi-level Phase Change Memory (PCM), a resistive memory, nanowire memory, ferroelectric transistor random access memory (FeTRAM), anti-ferroelectric memory, magnetoresistive random access memory (MRAM) memory that incorporates memristor technology, resistive memory including the metal oxide base, the oxygen vacancy base and the conductive bridge Random Access Memory (CB-RAM), or spin transfer torque (STT)-MRAM, a spintronic magnetic junction memory based device, a magnetic tunneling junction (MTJ) based device, a DW (Domain Wall) and SOT (Spin Orbit Transfer) based device, a thyristor based memory device, or a combination of any of the above, or other memory. The memory device may refer to the die itself and/or to a packaged memory product. In some embodiments, 3D crosspoint memory (e.g., Intel 3D XPoint™ memory) may comprise a transistor-less stackable cross point architecture in which memory cells sit at the intersection of word lines and bit lines and are individually addressable and in which bit storage is based on a change in bulk resistance.

(34) The processor **606** of the illustrative controller **602** may be embodied as, or otherwise include, any type of processor, controller, or other compute circuit capable of performing various tasks such as compute functions and/or controlling the functions of the work machine **100** and/or the fertilizer opener assembly **210**. For example, the processor **606** may be embodied as a single or multi-core processor(s), a microcontroller, or other processor or processing/controlling circuit. In some embodiments, the processor **606** may be embodied as, include, or otherwise be coupled to an FPGA, an application specific integrated circuit (ASIC), reconfigurable hardware or hardware circuitry, or other specialized hardware to facilitate performance of the functions described herein. Additionally, in some embodiments, the processor **606** may be embodied as, or otherwise include, a high-power processor, an accelerator co-processor, or a storage controller. In some embodiments still, the processor **606** may include more than one processor, controller, or compute circuit.

(35) The one or more frame actuator(s) **610** are each illustratively embodied as, or otherwise

include, any device or collection of devices capable of driving movement (e.g., translation and/or rotation) of the frame **110** in use of the work machine **100**. In some embodiments, the frame actuator(s) **610** may drive movement of the frame **110** relative to the chassis **104**. Additionally, in some embodiments, the frame actuator(s) **610** may drive movement of the frame **110** relative to the chassis **104** independently of movement of the cutting disk **212** (i.e., driven by the actuator **220**) relative to the soil.

(36) The one or more frame sensor(s) **612** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of measuring characteristics of position and/or movement of the frame **110** in use of the work machine **100**. In some embodiments, the frame sensor(s) **612** may be capable of measuring proximity and/or distance between the frame **110** and one or more components of the fertilizer opener assembly **210**. In such embodiments, the frame sensor(s) **612** may be capable of measuring the distance from a particular reference point on the frame **110** to the cutting disk **212** as the cutting disk **212** moves between the cutting and transport positions **200**, **300**.

(37) The one or more user input device(s) **614** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of providing input generated by a user to the controller **602** in use of the work machine **100**. In some embodiments, the input device(s) **614** may be provided separately from the dashboard **624**. Additionally, in some embodiments, the input device(s) **614** may be incorporated into, or otherwise form a portion of, the dashboard **624**.

(38) The one or more cutting implement position sensor(s) **616** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of measuring characteristics of position and/or movement of the cutting disk **212** and/or the gauge wheel **214** in use of the work machine **100**. The position sensor(s) **616** may measure the position of the cutting disk **212** and/or the gauge wheel **214** as the components **212**, **214** are driven by the actuator **220** for movement between the cutting position **200** and the transport position **300**.

(39) The one or more cutting implement force measurement sensor(s) **618** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of providing sensor data indicative of a force applied by the actuator **220** to the cutting disk **212** in use of the work machine **100**. The force measurement sensor(s) **618** may provide sensor data indicative of force(s) applied by the actuator **220** to the cutting disk **212** as the cutting disk **212** is driven by the actuator **220** for movement between the cutting position **200** and the transport position **300**. In some embodiments, the force measurement sensor(s) **618** may provide sensor data indicative of force(s) applied to the cutting disk **212** by the ground as the cutting disk **212** is driven by the actuator **220** for movement into, and out of, the soil.

(40) The one or more cutting implement actuator valve(s) **620** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of driving operation of the actuator **220** to cause movement of the cutting disk **212** and/or the gauge wheel **214** in use of the work machine **100**. In some embodiments, each of the actuator valve(s) **620** may be embodied as, or otherwise include, a solenoid valve capable receiving one or more control signals from a control device (e.g., the controller **602**) and selectively delivering fluid to one or more chambers of the actuator **220** to drive extension thereof in response to the control signals. In such embodiments, in response to one or more control signals issued from a control device, each solenoid valve may selectively evacuate fluid from one or more chambers of the actuator **220** to drive retraction of the actuator **220**.

(41) The one or more cutting implement actuator sensor(s) **622** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of measuring characteristics of position and/or movement of the actuator **220** in use of the work machine **100**. The actuator sensor(s) **622** may measure retraction and extension of the actuator **220** as the cutting disk **212** and/or the gauge wheel **214** are driven for movement by the actuator **220** between the cutting and transport positions **200**, **300**. Additionally, in some embodiments, the actuator sensor(s) **622** may

detect fluid delivery to, and fluid evacuation from, one or more chambers of the actuator **220**.

(42) The illustrative dashboard **624** includes a display **626** and a user interface **628** communicatively coupled thereto. The display **626** is configured to output or display various indications, messages, and/or prompts to an operator, which may be generated by the control system **600**. The user interface **628** is configured to provide various inputs to the control system **600** based on various actions, which may include actions performed by an operator.

(43) The one or more gauge wheel sensor(s) **630** are each illustratively embodied as, or otherwise include, any device or collection of devices capable of providing sensor data indicative of a force applied to the gauge wheel **214** in use of the machine **100**. In some embodiments, the force measured by the sensor(s) **630** may be indicative of a force applied to the gauge wheel **214** and/or the cutting disk **212** by the actuator **220**.

(44) Referring now to FIG. 7, in the illustrative embodiment, the controller **602** establishes an environment **700** during operation. The illustrative environment **700** includes a cutting implement positional adjustment module **702** and cutting depth control module **704**. Each of the modules, logic, and other components of the environment **700** may be embodied as hardware, firmware, software, or a combination thereof. As such, in some embodiments, one or more modules of the environment **700** may be embodied as circuitry or a collection of electrical devices. In such embodiments, one or more of the modules **702**, **704** may form a portion of the processor(s) **606** and/or other components of the controller **602**. Additionally, in some embodiments, one or more of the illustrative modules may form a portion of another module and/or one or more of the illustrative modules may be independent of one another. Further, in some embodiments, one or more of the modules of the environment **700** may be embodied as virtualized hardware components or emulated architecture, which may be established and maintained by the processor(s) **606** or other components of the controller **602**.

(45) The cutting implement positional adjustment module **702**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to selectively adjust the position and/or drive movement of the cutting disk **212** in use of the work machine **100**. To do so, in the illustrative embodiment, the cutting implement positional adjustment module **702** may perform one or more blocks of the method **800** described below with reference to FIGS. 8 and 9.

(46) The cutting depth control module **704**, which may be embodied as hardware, firmware, software, virtualized hardware, emulated architecture, and/or a combination thereof as discussed above, is configured to selectively adjust the depth and/or position of the cutting disk **212** in the soil when the cutting disk **212** is in the cutting position **200**. To do so, in the illustrative embodiment, the cutting depth control module **704** may perform one or more blocks of the method **800** described below with reference to FIGS. 8 and 9.

(47) Referring now to FIGS. 8 and 9, an illustrative method **800** of operating the work machine **100** may be embodied as, or otherwise include, a set of instructions that are executable by the control system **600** (e.g., the modules **702**, **704** of the controller **602**). The method **800** corresponds to, or is otherwise associated with, performance of the blocks described below in the illustrative sequence of FIGS. 8 and 9. It should be appreciated, however, that the method **800** may be performed in one or more sequences different from the illustrative sequence. Additionally, it should be appreciated that one or more blocks of the illustrative method **800** may be performed in parallel with, or contemporaneously with, one another.

(48) The illustrative method **800** begins with block **802**. In block **802**, the controller **602** receives input data from one or more devices. At least in some embodiments, the input data received by the controller **602** in block **802** may be provided by one or more of the following: the frame sensor(s) **612**, the user input device(s) **614**, the position sensor(s) **616**, the force measurement sensor(s) **618**, the sensor(s) **622**, or the dashboard **624** (e.g., the user interface **628**). Subsequent to block **802**, the method **800** proceeds to block **804**.

(49) In block **804** of the illustrative method **800**, the controller **602** selectively directs movement of the frame **110** (e.g., relative to the chassis **104**) based on the input data received in block **802**. Of course, in cases in which movement of the frame **110** is not desired based on the input data received in block **802**, performance of block **804** may be omitted. To perform block **804**, the controller **602** performs blocks **806** and **808**. In block **806**, the controller **602** provides control signals to the frame actuator(s) **610** to drive movement of the frame **110**. In block **808**, the controller **602** monitors the position and/or movement of the frame **110** based on input provided by the frame sensor(s) **612**. Following block **804**, the method **800** proceeds to block **810**.

(50) In block **810** of the illustrative method **800**, the controller **602** selectively directs movement of the cutting disk **212** independent of movement of the frame **110** based on the input data received in block **802**. Of course, in situations where movement of the cutting disk **212** is not desired based on the input data received in block **802**, performance of block **810** may be omitted. To perform block **810**, the controller **602** performs blocks **812** and **814**. In block **812**, the controller **602** provides control signals to the actuator **220** to drive movement of the cutting disk **212** independently of movement of the frame **110** relative to the chassis **104**. In block **814**, the controller **602** monitors the position and/or movement of the actuator **220** based on input provided by the actuator sensor(s) **622**. Subsequent to block **810**, the method **800** proceeds to block **816**.

(51) In block **816** of the illustrative method **800**, the controller **602** receives input from the position sensor(s) **616** indicative of the position of the cutting disk **212**. From block **816**, the method **800** subsequently proceeds to block **818**.

(52) In block **818** of the illustrative method **800**, the controller **602** determines the positional state of the cutting disk **212** based on the input received in block **816**. In some embodiments, if the controller **602** determines in block **818** that the cutting disk **212** is not in the cutting position **200**, and that movement of the cutting disk **212** toward the cutting position **200** is not desired (e.g., based on input provided by an operator), the controller **602** may omit performance of one or more remaining blocks of the method **800** discussed below. In the illustrative embodiment, however, presuming that movement of the cutting disk **212** toward the cutting position **200** is desired, the method **800** subsequently proceeds to block **920** from block **818**.

(53) In block **920** of the illustrative method **800**, the controller **602** directs movement of the cutting disk **212** to the cutting position **200**. It should be appreciated that to perform block **920**, at least in some embodiments, the controller **602** may issue control signals to the actuator valve(s) **620** to retract the actuator **220** and/or evacuate fluid from one or more chambers of the actuator **220**. Following block **920**, the method **800** subsequently proceeds to block **922**.

(54) In block **922** of the illustrative method **800**, the controller **602** receives input from the position sensor(s) **616** indicative of the position of the cutting disk **212** after movement of the cutting disk **212** to the cutting position **200** in block **920**. Subsequent to block **922**, the method **800** proceeds to block **924**.

(55) In block **924** of the illustrative method **800**, the controller **602** determines a depth of the cutting disk **212** in the soil based on the input received in block **922**. It should be appreciated that the depth may be determined by the controller **602** in block **924** based on a predetermined reference point and/or reference frame, such as the depth measured from the surface on which the wheels **106** are positioned, for example. In any case, subsequent to block **924**, the method **800** proceeds to block **926**.

(56) In block **926** of the illustrative method **800**, the controller **602** compares the depth determined in block **924** to a reference or predetermined depth. In some embodiments, the reference depth may be input to the controller **602** by an operator. Additionally, in some embodiments, the reference depth may be stored in the memory **604** of the controller **602** and retrieved by the processor **606** in order to perform block **926**. In any case, from block **926**, the method **800** subsequently proceeds to block **928**.

(57) In block **928** of the illustrative method **800**, the controller **602** adjusts the depth and/or position

of the cutting disk **212** based on the comparison performed in block **926**. It should be appreciated that in some situations, to perform block **928**, the controller **602** may provide control signals to the actuator **220** to retract the actuator **220** such that the cutting disk **212** moves deeper into the ground. Furthermore, it should be appreciated that in other situations, to perform block **928**, the controller **602** may provide control signals to the actuator **220** to extend the actuator **220** such that the cutting disk **212** moves away from the ground in the vertical direction **V** toward the transport position **300**. Regardless, subsequent to block **928**, the method **800** proceeds to block **930**.

(58) In block **930** of the illustrative method **800**, the controller **602** receives input from the force measurement sensor(s) **618** and/or the gauge wheel sensor(s) **630** indicative of a force (e.g., downforce) applied to the cutting disk **212** and/or the gauge wheel **214** by the actuator **220** through the positional adjustment linkage **230** when the cutting disk **212** is in the cutting position **200**. In some embodiments, the input received in block **930** may be indicative of force applied to the cutting disk **212** and/or the gauge wheel **214** through the positional adjustment arms **240**, **250** and the cutting implement guide link **260**. In any case, from block **930**, the method **800** subsequently proceeds to block **932**.

(59) In block **932** of the illustrative method **800**, the controller **602** determines the force applied to the cutting disk **212** and/or the gauge wheel **214** when the cutting disk **212** is in the cutting position **200** based on the input received in block **930**. Subsequent to block **932**, the method **800** proceeds to block **934**.

(60) In block **934** of the illustrative method **800**, the controller **602** compares the force determined in block **932** to a reference force. In some embodiments, the reference force may be input to the controller **602** by an operator. Additionally, in some embodiments, the reference force may be stored in the memory **604** of the controller **602** and retrieved by the processor **606** in order to perform block **934**. In any case, from block **934**, the method **800** subsequently proceeds to block **936**.

(61) In block **936** of the illustrative method **800**, based on the comparison performed in block **934**, the controller **602** selectively adjusts movement (i.e., retraction/extension) of the actuator **220** such that a relatively constant force is applied by the actuator **220** to the disk **212** and/or the gauge wheel **214** through the positional adjustment linkage **230** when the cutting disk **212** is in the cutting position **200**. It should be appreciated that in some embodiments, to perform block **936**, the controller **602** may receive input from the force measurement sensor(s) **618** similar to block **930**. In any case, at least in some embodiments, performance of block **936** corresponds to, or is otherwise associated with, performance of a single iteration of the method **800** by the controller **602**.

(62) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A work machine comprising: a chassis; a frame coupled to the chassis for movement relative thereto; and a fertilizer opener assembly coupled to the frame to selectively form trenches in the soil for the deposition of fertilizer, wherein: the fertilizer opener assembly includes an actuator that is extendable and retractable to vary a length of the actuator and a cutting implement coupled to the actuator, operation of the actuator drives movement of the cutting implement between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil, operation of the actuator drives movement of the cutting implement between the cutting position and the transport position independently of movement of the frame relative to the chassis, the fertilizer opener assembly includes a positional adjustment linkage coupled to the actuator and the cutting implement such

that operation of the actuator drives movement of the cutting implement through the positional adjustment linkage, the positional adjustment linkage includes a first positional adjustment arm that is directly connected to the actuator and coupled to the cutting implement and a second positional adjustment arm spaced from the first positional arm that is coupled to the actuator and to the cutting implement.

2. The work machine of claim 1, wherein extension of the actuator drives movement of the cutting implement toward the transport position, and wherein retraction of the actuator drives movement of the cutting implement toward the cutting position.

3. The work machine of claim 1, wherein the positional adjustment linkage includes a cutting implement guide link coupled to the cutting implement to at least partially define an angular orientation of the cutting implement relative to the soil, wherein the cutting implement guide link includes a first face and a second face arranged opposite the first face, and wherein the first positional adjustment arm contacts each of the first and second faces of the cutting implement guide link and the second positional adjustment arm contacts the second face of the cutting implement guide link.

4. The work machine of claim 3, wherein during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link at least partially cooperate to maintain a relatively constant angular orientation of the cutting implement relative to the soil.

5. The work machine of claim 3, wherein during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link at least partially cooperate to maintain a relatively constant upright orientation of the cutting implement guide link in which the cutting implement guide link extends away from the soil in a vertical direction.

6. The work machine of claim 3, wherein during movement of the cutting implement between the cutting position and the transport position, the first and second positional adjustment arms are arranged to extend parallel to one another.

7. The work machine of claim 1, wherein the fertilizer opener assembly includes a mount kit to attach the fertilizer opener assembly to the frame, wherein the mount kit includes a mount plate to contact the frame, a mount bracket extending outwardly from the mount plate that is affixed to the mount plate and to the actuator, a first mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm and the second positional adjustment arm at another end thereof opposite the one end, and a second mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm, the second positional adjustment arm, and the first mount link at another end thereof opposite the one end, and wherein the first and second positional adjustment arms are pivotally coupled to the first and second mount links to permit movement of the cutting implement between the cutting position and the transport position.

8. The work machine of claim 1, further comprising a control system coupled to the chassis, wherein the control system includes at least one sensor to provide sensor data indicative of a force applied by the actuator to the cutting implement and a controller communicatively coupled to the at least one sensor, wherein the controller includes memory having instructions stored therein and a processor coupled to the memory, and wherein the instructions are executable by the processor to cause the processor to receive the sensor data from the at least one sensor and to control operation of the actuator based on the sensor data such that a relatively constant force is applied by the actuator to the cutting implement when the cutting implement is in the cutting position.

9. The work machine of claim 8, wherein the instructions are executable by the processor to cause the processor to selectively retract the actuator to move the cutting implement into the soil when the cutting implement is in the cutting position, and wherein the instructions are executable by the processor to cause the processor to selectively extend the actuator to move the cutting implement in

a second direction opposite the first direction when the cutting implement is in the cutting position.

10. A fertilizer opener assembly for a work machine including a chassis and a frame coupled to the chassis for movement relative thereto, the fertilizer opener assembly comprising: an actuator that is extendable and retractable to vary a length of the actuator; a cutting implement to selectively form trenches in the soil for the deposition of fertilizer, wherein the cutting implement is movable relative to the soil between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil; a positional adjustment linkage coupled to the actuator and the cutting implement such that operation of the actuator drives movement of the cutting implement between the cutting position and the transport position through the positional adjustment linkage, and a control system including at least one sensor to provide sensor data indicative of a force applied by the actuator to the cutting implement and a controller communicatively coupled to the at least one sensor, wherein extension of the actuator drives movement of the cutting implement through the positional adjustment linkage toward the transport position, and wherein retraction of the actuator drives movement of the cutting implement through the positional adjustment linkage toward the cutting position.

11. The fertilizer opener assembly of claim 10, wherein the positional adjustment linkage includes a first positional adjustment arm that is directly connected to the actuator and coupled to the cutting implement and a second positional adjustment arm spaced from the first positional arm that is coupled to the actuator and to the cutting implement, and wherein during movement of the cutting implement between the cutting position and the transport position, the first and second positional adjustment arms are arranged to extend parallel to one another.

12. The fertilizer opener assembly of claim 11, wherein the positional adjustment linkage includes a cutting implement guide link coupled to the cutting implement to at least partially define an angular orientation of the cutting implement relative to the soil, wherein the cutting implement guide link includes a first face and a second face arranged opposite the first face, and wherein the first positional adjustment arm contacts each of the first and second faces of the cutting implement guide link and the second positional adjustment arm contacts the second face of the cutting implement guide link.

13. The fertilizer opener assembly of claim 12, wherein during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link at least partially cooperate to maintain a relatively constant angular orientation of the cutting implement relative to the soil.

14. The fertilizer opener assembly of claim 13, wherein during movement of the cutting implement between the cutting position and the transport position, the first positional adjustment arm, the second positional adjustment arm, and the cutting implement guide link at least partially cooperate to maintain a relatively constant upright orientation of the cutting implement guide link in which the cutting implement guide link extends away from the soil in a vertical direction.

15. The fertilizer opener assembly of claim 11, further comprising a mount kit to attach the fertilizer opener assembly to the frame, wherein the mount kit includes a mount plate to contact the frame, a mount bracket extending outwardly from the mount plate that is affixed to the mount plate and to the actuator, a first mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm and the second positional adjustment arm at another end thereof opposite the one end, and a second mount link that is affixed to the mount bracket at one end thereof and coupled to the first positional adjustment arm, the second positional adjustment arm, and the first mount link at another end thereof opposite the one end, and wherein the first and second positional adjustment arms are pivotally coupled to the first and second mount links to permit movement of the cutting implement between the cutting position and the transport position.

16. The fertilizer opener assembly of claim 10, wherein the controller includes memory having instructions stored therein and a processor coupled to the memory, and wherein the instructions are

executable by the processor to cause the processor to receive the sensor data from the at least one sensor and to control operation of the actuator based on the sensor data such that a relatively constant force is applied by the actuator to the cutting implement when the cutting implement is in the cutting position.

17. A method of operating a work machine including a chassis, a frame coupled to the chassis for movement relative thereto, and a fertilizer opener assembly coupled to the frame to selectively form trenches in the soil for the deposition of fertilizer, the method comprising: directing, by a controller, movement of the frame relative to the chassis; directing, by the controller, movement of a cutting implement of the fertilizer opener assembly independently of movement of the frame relative to the chassis, wherein directing movement of the cutting implement independently of movement of the frame relative to the chassis includes providing, by the controller, control signals to an actuator of the fertilizer opener assembly to cause extension or retraction of the actuator such that the cutting implement moves between a cutting position, in which the cutting implement makes contact with the soil to form trenches therein, and a transport position, in which the cutting implement is raised above the soil; directing, by the controller, movement of the cutting implement to the cutting position independently of movement of the frame relative to the chassis; and determining, by the controller, a depth of the cutting implement in the soil when the cutting implement is in the cutting position.

18. The method of claim 17, further comprising: comparing, by the controller, the determined depth to a reference depth; and adjusting, by the controller, a position of the cutting implement based on the comparison.

19. The method of claim 18, further comprising: determining, by the controller, a force applied to the cutting implement by the actuator when the cutting implement is in the cutting position; comparing, by the controller, the determined force to a reference force; and adjusting, by the controller, movement of the actuator based on the comparison such that a relatively constant force is applied by the actuator to the cutting implement when the cutting implement is in the cutting position.
